DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Course: B. Tech. **Branch : E & TC Engineering** **Semester : V**

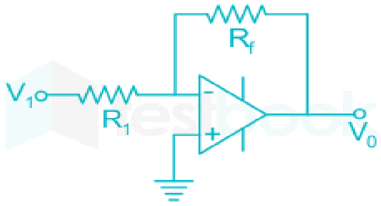
Subject Code & Name: BTETPE504A :: Analog Circuits

Max Marks: 60 **Date:08/02/2023** **Duration: 3Hr.**

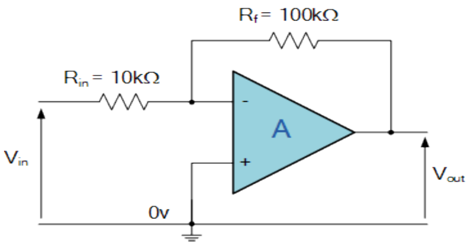
Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly in the answer sheet.*

		(Level/CO)	Marks
Q. 1	**Solve any *two* of the followings.**		
A)	Explain the block diagram of op-amp?	**Level 1**	**6**
B)	Define: i) CMRR ii) Gain iii) Slew rate	**Level 1**	**6**
C)	If V_{in}=2V, R_1= 200KΩ and R_f=500KΩ, find the output voltage of an Op-Amp.	**Level 2**	**6**



Q.2	**Solve any *two* of the followings.**		
A)	Find the closed loop gain of the following inverting amplifier circuit.	**Level 3**	**6**



B)	Discuss the application of an Op-Amp as a differentiator?	**Level 2**	**6**
C)	Explain Clippers and Clampers using Op-Amp.	**Level 1**	**6**
Q. 3	**Solve any *two* of the followings.**		
A)	With suitable diagrams, explain a summing amplifier using Op-amp.	**Level 1**	**6**

B)	Explain the concept of a non-inverting amplifier and virtual ground in Op-Amp.	**Level 1**	**6**
C)	Design a circuit using Op-amp that provides a gain of 5.5. Assume $R_{i=}$ 10kΩ, what value would you choose for R_f?	**Level 2**	**6**

Q.4 Solve any *two* of the followings.

A)	R = 10 k, v_1 = 2.011V, and v_2 = 2.017V. If RG is adjusted to 500 , determine: (a) The voltage gain, (b) The output voltage vo.	**Level 1**	**6**
B)	Define active filter and discuss its classification?	**Level 1**	**6**
C)	Explain the construction and working of a relaxation oscillator.	**Level 1**	**6**

Q. 5 Solve any *two* of the followings.

A)	Compare RC oscillators with LC oscillators.	**Level 1**	**6**
B)	Explain the working of Hartley oscillator with neat diagram.	**Level 2**	**6**
C)	Explain the concept of a weighted resistor. List is advantages, disadvantages.	**Level 1**	**6**

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